
125 kWp Grid-Connected Solar PV Installation — Ouarzazate, Morocco

PREFEASIBILITY ASSESSMENT · 25-YEAR DISCOUNTED CASHFLOW · MODEL v3.1 VALIDATED
NPV €936,184 · IRR 90.95% · Payback 1.1 yrs · LCOE €0.0236/kWh · 26.9× MOIC

SOLAR PV FEASIBILITY STUDY

Cooperative Agricole Drâa-Tafilalet

Cold Storage & Food Processing Facility
Ouarzazate, Morocco

125 kWp Grid-Connected Rooftop & Ground-Mounted System

Prefeasibility Assessment · 25-Year Discounted Cashflow Analysis

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Executive Summary

This study assesses the technical and financial viability of a 125 kWp rooftop and ground-mounted solar photovoltaic installation at the Cooperative Agricole Drâa-Tafilalet cold storage and food processing facility in Ouarzazate, Morocco. The facility currently sources 60% of its electricity from diesel generation at an effective cost of €0.35/kWh and 40% from the national grid at €0.12/kWh, resulting in a blended electricity cost of €0.26/kWh against an annual consumption of 220,000 kWh.

The objective of this study is to determine whether a solar PV installation can materially reduce this cost burden and diesel dependency over a 25-year project horizon.

Solar Resource

Ouarzazate benefits from one of the highest solar irradiation levels in Africa, with a BWh hot desert climate classification and fewer than 20 cloudy days per year. PVGIS analysis for the project coordinates (30.9335°N, 6.8937°W, altitude 1,140 m) returned an annual Global Horizontal Irradiance of 2,237 kWh/m²/year and an in-plane irradiation at 31° tilt of 2,486 kWh/m²/year — exceeding the European average by approximately 86%.

System Design and Energy Yield

The proposed system consists of 312 monocrystalline silicon panels rated at 400 Wp each, totalling 125 kWp of installed capacity. Energy yield simulation was performed using SAM (System Advisor Model) by NREL with the Ouarzazate TMY weather dataset, returning an annual energy production of 238,609 kWh in Year 1 (specific yield: 1,912 kWh/kWp/year). Cross-validation against PVGIS yielded 235,715 kWh/year — a difference of only 1.21%, confirming confidence in the production estimate.

Financial Performance

The total installed system cost is estimated at €68,800. A 25-year discounted cashflow analysis at a 5% real discount rate returns the following key metrics:

Year 1 Net Annual Savings	**€61,538**
Simple Payback Period	**1.1 years (13 months)**
Net Present Value (25 years)	**€936,184**
Internal Rate of Return	**90.9%**
25-Year Investment Multiple (MOIC)	**26.9x**
Levelized Cost of Energy	**€0.0145/kWh**

The LCOE of €0.0145/kWh compares favourably to the facility's current blended tariff of €0.26/kWh — a 94% reduction in electricity generation cost. The project generates a net financial benefit of €1,781,415 over the 25-year project lifetime.

Environmental Impact

The system displaces an estimated 193 tonnes of CO_2 equivalent in Year 1 — 124.6 tonnes from diesel displacement and 68.5 tonnes from grid electricity displacement. Over the 25-year lifetime, total CO_2 savings reach 4,549 tonnes, equivalent to permanently removing 34 passenger cars from the road annually.

Recommendation

Recommendation: Proceed

This assessment recommends proceeding with the proposed 125 kWp solar PV installation. The project demonstrates exceptional financial returns driven by the combination of Ouarzazate's world-class solar resource and the facility's elevated blended electricity tariff. Recommended next steps: (1) structural assessment of the installation area; (2) solicitation of at least three contractor quotes; (3) engagement with ONEE regarding grid connection and net metering arrangements.

1. Introduction

1.1 Project Objective

Solar energy is highly relevant for commercial facilities in Morocco due to the country's excellent solar resource and rising electricity prices. This feasibility study assesses the technical and financial viability of a grid-connected solar PV installation at a cold storage and food processing facility in Ouarzazate, Morocco, with the objective of reducing diesel consumption and total electricity cost over a 25-year project lifetime.

1.2 Methodology

Solar resource assessment was conducted using PVGIS (Photovoltaic Geographical Information System), developed by the European Commission Joint Research Centre. Detailed energy yield simulation was performed using SAM (System Advisor Model) by NREL, an industry-standard tool used by engineers and researchers globally. Financial analysis was conducted using a discounted cashflow model.

1.3 Scope and Limitations

This study constitutes a prefeasibility assessment. It does not include detailed structural analysis, electrical system design, grid connection assessment, battery storage analysis, or planning permission review. These elements would be addressed in a subsequent detailed design phase if the project proceeds.

2. Site and Resource Assessment

2.1 Site Description

The project site is located in Ouarzazate, Morocco, at coordinates 30.9335°N, 6.8937°W, at an altitude of approximately 1,140 metres above sea level. The facility is a cold storage and food processing operation with 800 m² of ground and roof area available for solar panel installation.

2.2 Solar Resource

Ouarzazate has a hot desert climate (BWh classification) with very high solar irradiation throughout the year. The region experiences long sunshine hours, clear skies, and low annual rainfall, making it one of the sunniest areas in Morocco — and indeed in Africa. The region receives approximately 2,100–2,500 kWh/m²/year of Global Horizontal Irradiance (GHI).

PVGIS analysis for the project location returned:

Parameter	Value
Annual Global Horizontal Irradiance (GHI)	**2,237 kWh/m²/year**
In-Plane Irradiation (31° tilt, south-facing)	**2,486 kWh/m²/year**
Climate Classification	**BWh — Hot Desert**
Average Cloudy Days per Year	**< 20 days**

Monthly irradiation data is presented in Figure 1 of the supporting annexes. System losses based on the PVGIS simulation include angle of incidence, spectral effects, temperature and low irradiance losses, as presented in Figure 2.

3. System Design

3.1 System Overview

The proposed system consists of a 125 kWp grid-connected ground and roof solar photovoltaic installation. The system comprises 312 solar PV modules rated at 400 Wp each, configured as 24 modules per string, 13 strings in parallel in 1 subarray. The system feeds electricity directly into the facility's internal electrical network, offsetting grid electricity purchases.

3.2 Key Components

Component	Specification	Quantity
Solar PV Module	400 Wp, 20% efficiency, monocrystalline silicon	312
Inverter	Yaskawa Solectria XGI 1500-125/125-3S [600V]	1
DC System Size	125 kWp	—
Roof Area Required	800 m² (at 2 m² per panel)	—
Mounting System	Ground and rack mounted	—
Panel Orientation	South-facing, 0° azimuth	—

3.3 System Losses

The simulation incorporates total system losses of approximately 11.4% (as returned by SAM), accounting for: soiling (5%), wiring losses (1%), inverter efficiency losses, temperature-related losses, and system availability. These loss assumptions are consistent with industry standards for commercial PV installations in arid climates.

4. Energy Yield Analysis

4.1 Simulation Results

Energy yield simulation was performed using SAM with the Ouarzazate TMY (Typical Meteorological Year) weather dataset. The simulation returned:

Energy Yield Metric	Value
Annual Energy Production — Year 1 (SAM)	**238,609 kWh**
Specific Yield	**1,912 kWh/kWp/year**
System Capacity Factor	**21.8%**
Performance Ratio	**81%**
Cross-Check — PVGIS Estimate	**235,715 kWh/year**
SAM vs PVGIS Divergence	**1.21%**

The system capacity factor of 21.8% significantly exceeds the global average of 14–18% for fixed-tilt commercial systems, reflecting the superior solar resource at this location. Monthly production data is presented in Figure 3. Peak production occurs in March and April at approximately 22,214 kWh per month.

4.2 Cross-Validation

As a validation step, SAM results were cross-checked against PVGIS. The 1.21% divergence between independent simulation tools is well within the accepted engineering tolerance of ±5%, providing strong confidence in the production estimate.

5. Financial Analysis

5.1 Capital Cost Assumptions

The total installed system cost is estimated at €68,800, based on current market rates for commercial rooftop solar installations in Morocco. This includes module supply, inverter, mounting hardware, electrical balance of system, installation labor, and engineering and permitting costs. Annual operations and maintenance costs are estimated at €500 per year.

Cost Item	Cost (€)
Solar PV Modules	€46,800
Inverter	€5,000
Balance of System	€8,000
Installation Labor	€6,000
Engineering and permitting	€3,000
Total System Cost	**€68,800**

5.2 Financial Assumptions

The analysis applies the following base-case assumptions:

Current Blended Electricity Tariff	**€0.26/kWh**
Annual Electricity Price Escalation	**2.0% per year**
Real Discount Rate	**5.0%**
Annual System Degradation	**0.5% per year**
Annual O&M Cost	**€500/year**
Project Lifetime	**25 years**

5.3 Financial Results

The system generates estimated annual electricity savings of €62,038 in Year 1, rising to €88,475 by Year 25 as electricity prices increase. Over the 25-year project lifetime, cumulative electricity savings total €1,850,215. After deducting the initial investment and maintenance costs, the net financial benefit to the facility is €1,781,415.

Year 1 Net Annual Savings	**€61,538**
Simple Payback Period	**1.1 years (13 months)**
Net Present Value (25 years)	**€936,184**
Internal Rate of Return (IRR)	**90.9%**
25-Year Cumulative Net Benefit	**€1,781,415**
Levelized Cost of Energy (LCOE)	**€0.0145/kWh**
25-Year Investment Multiple (MOIC)	**26.9x**

Figure 4 in the supporting annexes illustrates the cumulative net benefit over the project lifetime, showing the point at which cumulative savings exceed the initial investment.

5.4 Sensitivity Note

The financial results are sensitive to electricity price assumptions. If electricity prices rise faster than the assumed 2% per year, the payback period shortens and overall returns improve. The analysis does not include potential revenue from electricity export to the grid, which could further improve financial performance if the facility enters a net metering or feed-in arrangement with ONEE.

6. Environmental Analysis

The 125 kWp solar PV system is estimated to produce 238,609 kWh of clean electricity in its first year of operation. Based on Morocco's current grid emission factor of 0.718 kg CO_2/kWh and a diesel emission factor of 2.68 kg CO_2/litre, this displaces approximately 193 tonnes of CO_2 equivalent per year.

Annual CO_2 Savings — Grid Displacement	**68.5 tonnes CO_2**
Annual CO_2 Savings — Diesel Displacement	**124.6 tonnes CO_2**
Total Annual CO_2 Savings (Year 1)	**193 tonnes CO_2**
25-Year Lifetime CO_2 Savings	**4549 tonnes CO_2**
Equivalent Passenger Cars Removed Annually	**94 cars**

Beyond direct emissions savings, diesel displacement reduces the facility's exposure to fuel supply chain risk and generator maintenance costs — operational benefits not captured in the financial model but representing additional economic value.

7. Conclusions and Recommendations

7.1 Technical Conclusion

The solar resource assessment confirms that Ouarzazate offers excellent conditions for solar PV generation, with annual in-plane irradiation of approximately 2,486 kWh/m²/year. The proposed 125 kWp system is technically well-suited to the site and is expected to produce approximately 238,609 kWh annually, achieving a performance ratio of 81% and a capacity factor of 21.8%.

7.2 Financial Conclusion

The financial analysis demonstrates that the proposed solar installation represents a sound investment for the cold storage and food processing facility. With a payback period of 1.1 years (13 months) and a positive net present value of €936,184 over 25 years, the project delivers strong economic returns. The LCOE of €0.0145/kWh represents a 94% reduction relative to the current blended electricity tariff.

7.3 Recommended Next Steps

Should the client decide to proceed, the following steps are recommended:

1.	Commission a structural assessment of the roof and ground area to confirm load-bearing capacity for panel and mounting system weight.
2.	Obtain quotes from at least three certified solar installation contractors to validate cost assumptions.
3.	Engage with ONEE regarding grid connection requirements and any available net metering or feed-in arrangements.
4.	Review applicable planning permissions and building regulations with the local municipality (Commune de Ouarzazate).

7.4 Study Limitations

This study is a prefeasibility assessment based on modelled data and standard industry assumptions. Actual system performance may differ from modelled results due to site-specific factors not captured in this analysis. A detailed engineering study is recommended before a final investment decision.

References

[1] European Commission Joint Research Centre. PVGIS (Photovoltaic Geographical Information System). Available at: re.jrc.ec.europa.eu/pvg_tools

[2] National Renewable Energy Laboratory. System Advisor Model (SAM). Available at: sam.nrel.gov

[3] Office National de l'Electricité et de l'Eau Potable (ONEE). Moroccan grid emission factor and electricity tariff data.